

CLOCK GATING

For The Internet Of Things



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The Internet of Things (IoT) connects smartphones, desktop systems, smart sensors, smart automobiles, and the things embedded with electronics like software, sensors and actuators with the Internet. With rapid advances in fields like software, semiconductor and communication, hardware miniaturisation has pushed the incorporation of intelligence and connectivity into the smallest of things like sensors in clothes, to huge systems like automobiles, factories or even cities. Analysts estimate that there will be 50 billion Internet-connected devices by 2020, up from 25 billion in 2015. This predicted explosion of IoT devices affects various evolving and growing markets as well as entirely new applications.

This level of integration of intelligence and connectivity requires highly-reliable, highly-secure, high-performance, low-cost and low-power system boards and integrated chips. Despite the variety of target applications, all such devices demand low energy/power consumption, high reliability, connectivity, interoperability, security and privacy.

Further, while meeting these demands, time-to-market is a critical metric that determines whether an IoT product will capture market share and, thus, the greatest revenue opportunity. Hence, a design methodology that meets design goals while ensuring fast time-to-market is crucial.

The IoT encompasses technologies like machine-to-machine (M2M), machine-to-infrastructure, machine-to-environment, the Internet of everything, the Internet of intelligent things and intelligent systems, among others. Cost of IoT products plays a disproportionately important role in market viability. Since most IoT applications are driven to provide cost-effective solutions, IoT system developers do not target newer evolving protocols or technology nodes.

Battery-backed-up devices have gained more popularity in the last few

years. This demands smaller form factor, lower weight, increased battery life and high performance. Architects and designers have learned that there is no single solution to the problem—power optimisation needs to be considered at all phases of the design life cycle, including architecture, algorithms, technology, register transfer level (RTL) design, backend activities and so on.

Low-power techniques

Power dissipation is a concern while designing high-performance systems. This makes clock signals a great source of power dissipation. High-frequency systems use a large number of clock pulses. Consider a system that is fed clock pulses to change its output. In most cases, pulses go waste as output does not change. Thus, extra clock pulses form a part of power dissipation rather than adding to circuit performance. Thus, there is a need for clock gating technique, which can be applied to reduce power.

It is better to consider power in all stages of the system: software and hardware design, irrespective of whether it is a system board design or a chip design. For integrated circuit (IC) design flow, there are several stages in which low-power design techniques can be used. These include requirement specification, architecture design, RTL development, synthesis activities and physical design activities.

Architectural level

At the architectural level, designers make decisions such as choosing clock frequency, micro-architecture of design, partitioning the design with respect to clock and so on. In this phase, they do not have a lot of flexibility to reduce overall power consumption.

Following are three architectural low-power techniques used by designers:

1. Clock gating
2. Architectural clock gating
3. Dynamic frequency variation

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